

Course: ECEN 602 Computer Communication and Networking

Assignment: Machine Problem 1 — TCP Echo Server and Client

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If you get the error message: Failed to create home directory
Then you will need to first create your home directory. Directions are available in your browser by visiting the following web page: <https://docs.ao.tamu.edu/docs/endpoints/engr-home-directory>

```
COMPUTER NAME: olympus-login.engr.tamu.edu

[shihyuyang@olympus-login ~]$ cd ~/ecen-602-mp
[shihyuyang@olympus-login ecen-602-mp]$ cd MP1
[shihyuyang@olympus-login MP1]$ ls -la
total 328
drwx----- 4 shihyuyang nobody   249 Sep 20 17:51 .
drwx----- 4 shihyuyang nobody   129 Sep 20 17:50 ..
-rwx----- 1 shihyuyang nobody  2027 Sep 20 14:01 Makefile
-rwx----- 1 shihyuyang nobody  2232 Sep 20 14:01 README.md
drwx----- 3 shihyuyang nobody    85 Sep 20 14:01 docs
-rwx----- 1 shihyuyang nobody  2218 Sep 20 14:01 echo.c
-rwx----- 1 shihyuyang nobody  2780 Sep 20 14:01 echo_io.c
-rwx----- 1 shihyuyang nobody  3248 Sep 20 14:01 echo_io.h
-rwx----- 1 shihyuyang nobody 11303 Sep 20 14:01 echos.c
drwx----- 2 shihyuyang nobody    64 Sep 20 14:01 tests
-rwx----- 1 shihyuyang nobody  2672 Sep 20 14:01 util.c
-rwx----- 1 shihyuyang nobody  1149 Sep 20 14:01 util.h
[shihyuyang@olympus-login MP1]$ make test
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -c -o echos.o echos.c
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -c -o echo_io.o echo_io.c
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -c -o util.o util.c
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -o echos.o echo_io.o util.o
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -l. -o tests/test_echos tests/test_echos.c
./tests/test_echos --server ./echos
starting ./echos on port 43891
PASS 1. line terminated by newline [17 bytes round-tripped]
PASS 1b. multiple lines, one connection [4 lines in order]
PASS 2. max-length line, no newline [4096 bytes echoed without a newline]
PASS 2b. line longer than the buffer [6152 bytes preserved across reads]
PASS 3. no characters, then EOF [empty stream closes cleanly; bare newline echoes]
PASS 4. client killed after sending text [RST handled, server still serving]
PASS 5. three simultaneous clients [3 concurrent connections, no cross-talk]
PASS 6. no zombie children after 12 connections [SIGCHLD handler reaped every child]
PASS 7. line split across TCP segments [readline() waited for the newline]
PASS 8. 8-bit clean (NUL and high bytes) [writen() used the length, not strlen()]
PASS 9. 48 rapid sequential connections [accept/fork loop stable]

server shut down cleanly on SIGINT

11/11 cases passed
[shihyuyang@olympus-login MP1]$ █
```

Ran test_echos.c; all test cases passed.

1. line of text terminated by a newline

If you get the error message: Failed to create home directory
Then you will need to first create your home directory. Directions are available in your browser by visiting the following web page: <https://docs.ao.tamu.edu/docs/endpoints/engr-home-directory>

```
COMPUTER NAME: olympus-login.engr.tamu.edu

[shihyuyang@olympus-login ~]$ cd ~/ecen-602-mp/MP1
-bash: cd: /home/grads/s/shihyuyang/ecen-602-mp/MP1: No such file or directory
[shihyuyang@olympus-login ~]$ ./echo 127.0.0.1 15000
-bash: ./echo: No such file or directory
[shihyuyang@olympus-login ~]$ cd ~/ecen-602-mp/MP1
[shihyuyang@olympus-login MP1]$ ./echo 127.0.0.1 15000
Hi this is Aloha speaking
Hi this is Aloha speaking
Hi
Hi
█

[shihyuyang@olympus-login MP1]$ ./echos 15000
[2026-09-20 18:01:05] echos: listening on 0.0.0.0:15000 (pid 3672023)
./echo 127.0.0.1 15000
^C[2026-09-20 18:02:38] echos: shutdown requested, closing listening socket
[shihyuyang@olympus-login MP1]$ make
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -c -o echo.o echo.c
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -o echo echo.o echo_io.o util.o
[shihyuyang@olympus-login MP1]$ ./echos 15000
[2026-09-20 18:03:56] echos: listening on 0.0.0.0:15000 (pid 3708245)
[2026-09-20 18:05:39] 127.0.0.1:38116: accepted, handed to child pid 3731290
[2026-09-20 18:05:39] 127.0.0.1:38116: connection established (child pid 3731290)
[2026-09-20 18:06:41] 127.0.0.1:38116: echoed 26 bytes
[2026-09-20 18:07:08] 127.0.0.1:38116: echoed 3 bytes
█
```

Client sent two different lines of text and received the echoed responses from the server.

2. Max line length without a newline

```
[shihyuyang@oilympus-login MP1]$ ./echos 15000
[2026-09-20 18:03:56] echos: listening on 0.0.0.0:15000 (pid 3708245)
[2026-09-20 18:05:39] 127.0.0.1:38116: accepted, handed to child pid 3731290
[2026-09-20 18:05:39] 127.0.0.1:38116: connection established (child pid 3731290)
[2026-09-20 18:06:41] 127.0.0.1:38116: echoed 26 bytes
[2026-09-20 18:07:08] 127.0.0.1:38116: echoed 3 bytes
[2026-09-20 18:12:01] 127.0.0.1:38116: EOF from client
[2026-09-20 18:12:01] 127.0.0.1:38116: connection closed (child pid 3731290 exiting)
[2026-09-20 18:18:01] 127.0.0.1:42990: accepted, handed to child pid 3884768
[2026-09-20 18:18:01] 127.0.0.1:42990: connection established (child pid 3884768)
[2026-09-20 18:18:16] 127.0.0.1:42990: EOF from client
[2026-09-20 18:18:16] 127.0.0.1:42990: connection closed (child pid 3884768 exiting)
[2026-09-20 18:31:10] 127.0.0.1:56268: accepted, handed to child pid 4055598
[2026-09-20 18:31:10] 127.0.0.1:56268: connection established (child pid 4055598)
[2026-09-20 18:31:10] 127.0.0.1:56268: echoed 4096 bytes (no newline: line limit reached)
[2026-09-20 18:31:10] 127.0.0.1:56268: EOF from client
[2026-09-20 18:31:10] 127.0.0.1:56268: connection closed (child pid 4055598 exiting)
```

3. Line with no characters and EOF

```
[shihyuyang@olympus-login MP1]$ ./echo 127.0.0.1 15000
[shihyuyang@olympus-login MP1]$
```

```

[shihyuyang@olympus-login MP1]$ ./echos 15000
[2026-09-20 18:01:05] echos: listening on 0.0.0.0:15000 (pid 3672023)
./echo 127.0.0.1 15000
^C[2026-09-20 18:02:38] echos: shutdown requested, closing listening socket
[shihyuyang@olympus-login MP1]$ make
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -c -o echo.o echo.c
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -o echo echo.o echo_io.o util.o
[shihyuyang@olympus-login MP1]$ ./echos 15000
[2026-09-20 18:03:56] echos: listening on 0.0.0.0:15000 (pid 3708245)
[2026-09-20 18:05:39] 127.0.0.1:38116: accepted, handed to child pid 3731290
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[2026-09-20 18:12:01] 127.0.0.1:38116: connection closed (child pid 3731290 exiting)
[2026-09-20 18:18:01] 127.0.0.1:42990: accepted, handed to child pid 3884768
[2026-09-20 18:18:01] 127.0.0.1:42990: connection established (child pid 3884768)
[2026-09-20 18:18:16] 127.0.0.1:42990: EOF from client
[2026-09-20 18:18:16] 127.0.0.1:42990: connection closed (child pid 3884768 exiting)

```

The client sent EOF without any text; the server closed connection immediately without error.

4. client terminated after entering text

```

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[shihyuyang@olympus-login ~]$ ./echo 127.0.0.1 15000
-bash: ./echo: No such file or directory
[shihyuyang@olympus-login ~]$ cd ~/ecen-602-mp/MP1
[shihyuyang@olympus-login MP1]$ ./echo 127.0.0.1 15000
Hi this is Aloha speaking
Hi this is Aloha speaking
Hi
Hi
[shihyuyang@olympus-login MP1]$ █

-----
[shihyuyang@olympus-login MP1]$ ./echos 15000
[2026-09-20 18:01:05] echos: listening on 0.0.0.0:15000 (pid 3672023)
./echo 127.0.0.1 15000
^C[2026-09-20 18:02:38] echos: shutdown requested, closing listening socket
[shihyuyang@olympus-login MP1]$ make
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -c -o echo.o echo.c
gcc -std=c11 -D_DEFAULT_SOURCE -Wall -Wextra -Wpedantic -O2 -g -o echo echo.o echo_io.o util.o
[shihyuyang@olympus-login MP1]$ ./echos 15000
[2026-09-20 18:03:56] echos: listening on 0.0.0.0:15000 (pid 3708245)
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[2026-09-20 18:07:08] 127.0.0.1:38116: echoed 3 bytes
[2026-09-20 18:12:01] 127.0.0.1:38116: EOF from client
[2026-09-20 18:12:01] 127.0.0.1:38116: connection closed (child pid 3731290 exiting)

```

Client sent 2 lines of text and terminated via EOF (Ctrl+D); server closed connection.

Three clients connected to the server

Executed `./tests/three_clients.sh` to run three concurrent client connections simultaneously. The server parent successfully forked three child processes (PIDs 4142739, 4142756, 4142779) to serve each client in parallel.

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